

Supervised Machine Learning

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Machines don't just wake up knowing how to recognize photos or predict your next favorite song. They have to be trained first. Have you ever wondered why we have to teach machines first and then ask them to give us answers later?

Can't we, as humans, just predict things ourselves?

Well, yes, we can, but only up to a point. Humans might be able to predict outcomes with around 50% accuracy, while machines, when trained properly, can often reach 90% or more. The real difference isn't intelligence; it's speed and scale. Machines can process massive amounts of data much faster than we ever could.

But teaching a machine isn't simple. It's not like explaining something to a child once and expecting it to understand. We need to follow certain rules, prepare the data carefully, and make sure the information we feed it is meaningful and consistent.

That's where machine learning comes in a field dedicated to training computers to recognize patterns and make predictions based on data. And depending on how that data is structured, we have two main ways to train these models:

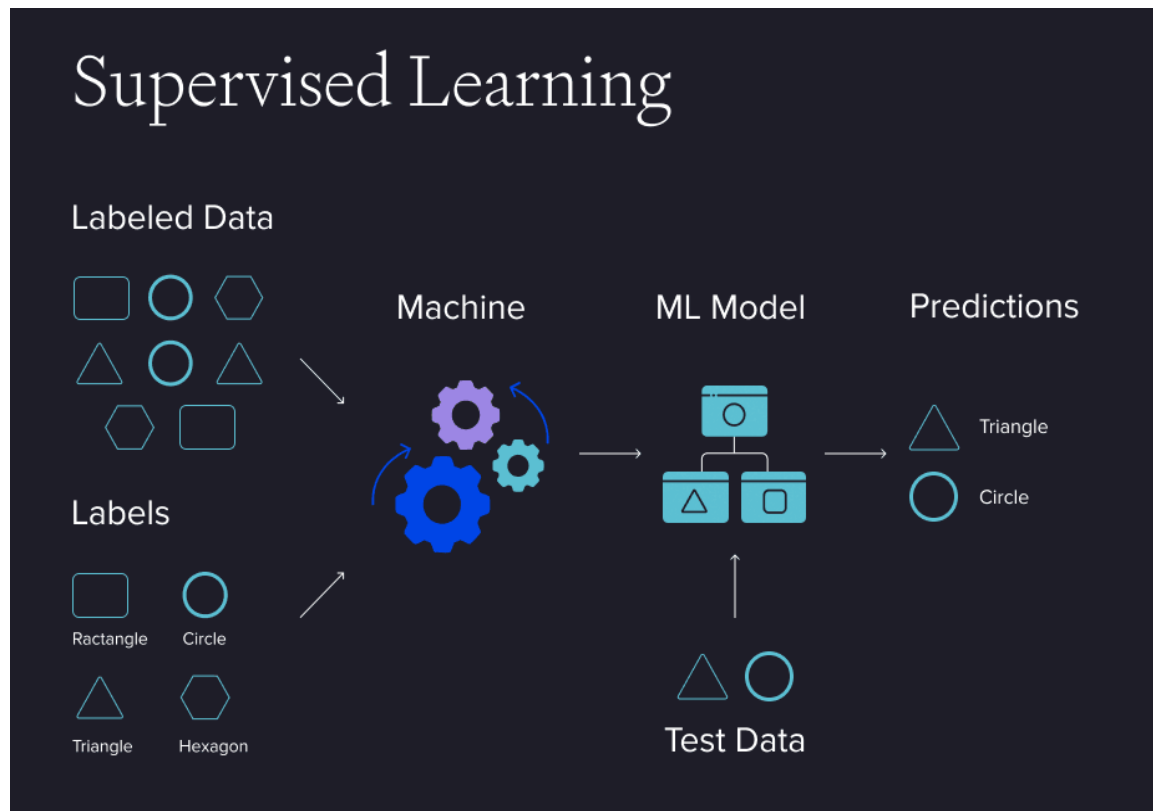
1. Supervised Machine Learning
2. Unsupervised Machine Learning

In this article we will explore about Supervised Machine Learning

Supervised Machine Learning:

Supervised learning is a type of machine learning where algorithms are trained using labelled data. The model learns to associate inputs with the correct outputs by analysing examples in the dataset. These examples consist of input-output pairs, where the input is a set of features (such as variables or attributes), and the output is the corresponding label or value. The main objective of supervised machine learning is to enable the model to generalise from these

examples and generate precise forecasts on new, unseen data. In supervised machine learning, the word “**labeled**” simply means that **each piece of data already has the correct answer attached to it**.



Types of Supervised Learning

There are two types of supervised Learning.

1. Classification
2. Regression

Alright, before we get into classification and regression, let's quickly talk about **training sets** and **test sets** you'll hear these terms everywhere.

So here's the idea behind supervised learning: you're teaching a machine to make predictions using data where you already know the answers. But there's a problem how do you know if it actually learned something, or if it just memorized everything like a parrot?

That's where splitting comes in.

Most of the time, we will use around **70–80%** of data for **training** (that's where the learning happens), and we set aside the rest maybe **20–30%** for **testing**. It's honestly like school. Training data

is the stuff we study from, and our test data is the final exam. we don't get to see the exam questions while studying, right? Same logic here. But okay, before split anything, we need to know *what* we're trying to predict. Every dataset has something we care about we call that the target variable. And then there are all these other things, the features, that might help us figure it out.

Let me make this concrete. Say you're working on a loan approval system. Your data might have someone's age, their salary, what kind of job they have, how many kids they're supporting — stuff like that. And what you want to know is: should we approve this person's loan or not? That yes-or-no answer? That's your target. Everything else the age, salary, dependents those are the features feeding into it.

Classification:

If the target variable is a **category** or represents a **discrete value**, then the problem is known as a classification problem.

In simple terms, classification is about **sorting data into groups**. The model looks at the input features and decides which category each data point belongs to.

For example, a classification model might predict:

- Whether an email is *spam* or *not spam*
- Whether a loan will be *approved* or *rejected*
- Whether a tumor is *benign* or *malignant*

Common Algorithms Used for Classification:

- K-Nearest Neighbors (KNN)
- Naive Bayes
- Decision Tree
- Logistic Regression

Regression:

Now, not every prediction is about categories. Sometimes, we want to predict a number something that changes continuously rather than fitting into a box.

That's where regression comes in.

In simple terms, regression is all about **predicting values**.

Instead of saying *yes* or *no*, the model tries to estimate *how*

much, how many, or how likely.

Think of a few examples:

- Predicting the price of a house based on its size, location, and number of rooms
- Estimating a person's salary using their education and experience
- Forecasting sales or temperature for the next month

In each of these cases, the output is a continuous value, not a fixed label.

The model learns from past data to find relationships between the input features (like square footage or years of experience) and the target variable (like price or salary).

Common Algorithms Used for Regression:

- Linear Regression
- Polynomial Regression
- Decision Tree Regressor
- Random Forest Regressor

Before we use supervised learning

There are some requirements to satisfy before performing supervised learning.

- Our data must not have missing values.
- Our data must be in numeric format and stored as pandas DataFrames or Series, or NumPy arrays. This requires some exploratory data analysis first to ensure data is in the correct format. Various pandas methods for descriptive statistics, along with appropriate data visualizations, are useful in this step.